Microstructure Stylized Facts

IEOR E4576 — Data-Driven Models in Finance

Order Flow & Arrival

1. **Power-law limit order placement.** $P(\Delta) \sim (\Delta + \Delta_1)^{-\mu}$, $\mu \approx 0.6$.
2. **Power-law order sizes.** $P(V > x) \sim x^{-1-\alpha}$, $\alpha \approx 1.5$.
3. **Long memory in order flow.** $\mathrm{Corr}(\epsilon_t, \epsilon_{t+\tau}) \sim \tau^{-\beta}$, $\beta \approx 0.5$–0.7. Attributed to institutional order splitting.
4. **Order cancellation rates.** Lifetime increases roughly exponentially with distance from best quote.

Order Book Shape

5. **Universal hump-shaped LOB.** Low volume at best quote, maximum a few ticks away, then decay. Approximately universal across stocks after rescaling.
6. **Log-volume is Gamma-distributed.** $\log_{10}(V)$ at each price level follows a Gamma distribution, shape $\gamma \approx 0.7$.

Price Impact

7. **Square-root impact law.** $\Delta P \sim \sigma\sqrt{Q/V}$. Holds across asset classes, time periods, and order sizes.
8. **Temporary vs. permanent impact.** 50–70% of peak meta-order impact is permanent; 30–50% reverts over minutes to hours.

Returns & Volatility

9. **Volatility clustering.** $\mathrm{Corr}(|r_t|, |r_{t+\tau}|) \sim \tau^{-\gamma}$, $\gamma \approx 0.2$–0.4.
10. **Heavy-tailed returns (inverse cubic law).** $P(|r| > x) \sim x^{-\alpha}$, $\alpha \approx 3$.
11. **No return autocorrelation.** $\mathrm{Corr}(r_t, r_{t+\tau}) \approx 0$ for $\tau \gtrsim 1$ min. Negative at tick-level due to bid-ask bounce.
12. **Epps effect.** Measured cross-asset correlation decreases as sampling frequency increases, due to asynchronous trading and microstructure noise.

Spread & Tick Size

13. **Spread scales with volatility, inversely with volume.** $s \propto \sigma$, $s \propto 1/V$.
14. **Tick-size clustering.** When tick size is not binding, prices cluster on round numbers. When binding, top-of-book becomes a queue-priority game.

Imbalance & Short-Term Predictability

15. **Book imbalance predicts short-term returns.** $I = (V_{\text{bid}} - V_{\text{ask}})/(V_{\text{bid}} + V_{\text{ask}})$ is positively correlated with subsequent mid-price moves over seconds to minutes.
16. **Trade imbalance predicts returns.** $\Delta P = \lambda \cdot (\text{net order flow}) + \text{noise}$ (Kyle's lambda).

Intraday Patterns

17. **U-shaped intraday volume and volatility.** High at open, low midday, high at close. Spread follows inverted pattern.

Fact	**Form**	**Reference**
LO placement	$P(\Delta) \sim \Delta^{-0.6}$	Bouchaud, Mézard, Potters (2002)
Order sizes	$P(V > x) \sim x^{-2.5}$	Gopikrishnan et al. (2000)
Order flow memory	$C(\tau) \sim \tau^{-0.6}$	Lillo & Farmer (2004)
LOB shape	Hump, universal after rescaling	Bouchaud, Mézard, Potters (2002)
Meta-order impact	$\Delta P \sim \sigma\sqrt{Q/V}$	Tóth et al. (2011)
Impact decay	50–70% permanent	Farmer et al. (2013)
Return tails	$P(\|r\| > x) \sim x^{-3}$	Gopikrishnan et al. (1999)
Vol clustering	$\mathrm{Corr}(\|r_t\|, \|r_{t+\tau}\|) \sim \tau^{-0.3}$	Cont (2001)
Return autocorr	≈ 0 beyond seconds	Cont (2001)
Epps effect	Corr $\downarrow$ as freq $\uparrow$	Epps (1979)
Spread	$\propto \sigma$, $\propto 1/V$	Roll (1984)
Book imbalance	I predicts ΔP_{mid}	Cont, Kukanov, Stoikov (2014)
Intraday patterns	U-shaped vol & volume	Wood, McInish, Ord (1985)